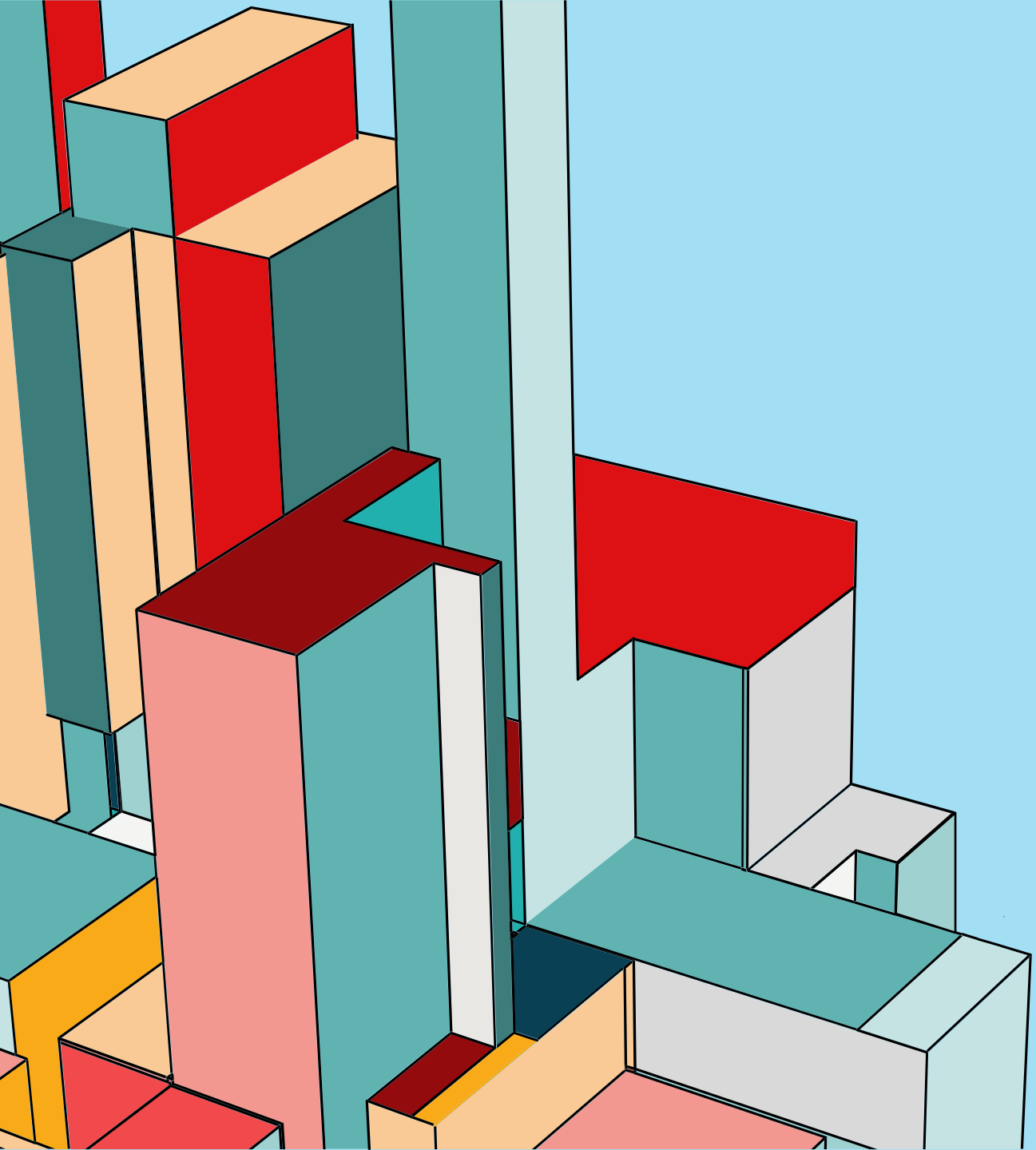


APPLIED DATA SCIENCE PROJECT

SYNTHETIC DATASET GENERATION FOR CAR-WASH HAZARD DETECTION

Politecnico di Torino
FONDAZIONE links PASSION FOR INNOVATION
ellis European Laboratory for Learning and Intelligent Systems

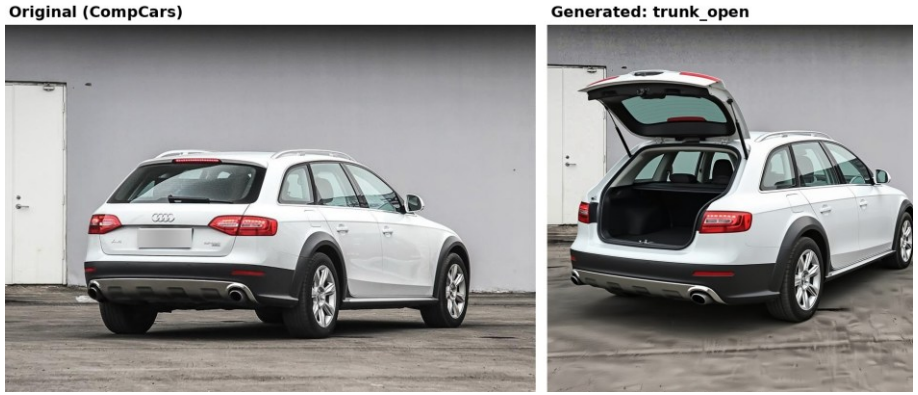
PROJECT OVERVIEW

The problem

- Open doors, roof boxes and bike racks can jam or damage an automatic car wash.
- A camera at the entrance could spot them, but there is no real footage to train it on.
- Generated images can fill the gap, but only if they are realistic and correctly labelled.

The goal

- Define and **automate** a pipeline that **generates hazard images, validates each one against hallucinations and labels it with SAM 3**[2].
- It runs end to end without manual steps; people only review a sample.
- The output is a labelled dataset that a hazard detector can later be trained on.

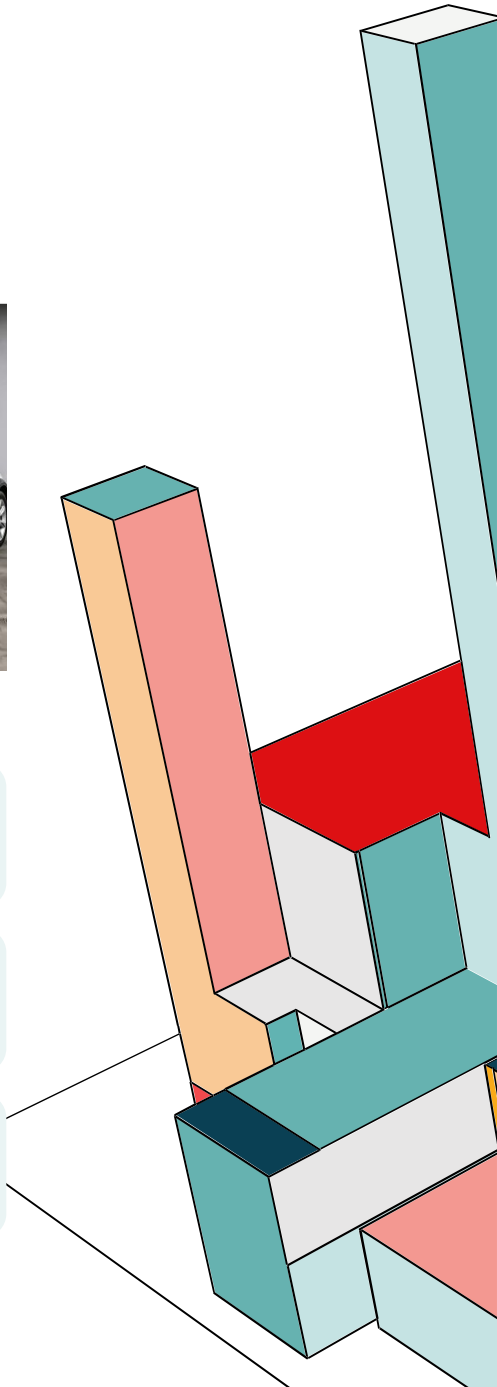


What students deliver

1 Generation pipeline
FLUX.1 Kontext[1] adds one hazard per photo

2 Hallucination checks
Reject images where the car changed

3 SAM 3[2] labelling
One hazard mask and class per image



1 Black Forest Labs, FLUX.1 Kontext: Flow Matching for In-Context Image Generation and Editing, arXiv:2506.15742, 2025.
2 Carion et al. (Meta), SAM 3: Segment Anything with Concepts, arXiv:2511.16719, 2025.

DATA

Dataset

- CompCars (CUHK, CVPR 2015): web and surveillance photos, JPG, about 20 GB.
- Labels: viewpoint, car bounding box, make, model and body type.
- Download: mmlab.ie.cuhk.edu.hk/datasets/comp_cars

How it's created

- CompCars is a fixed academic release that nobody updates.
- Students generate the hazard images themselves, editing each source photo from a text prompt with an image-editing model, for example FLUX.1 Kontext[1].

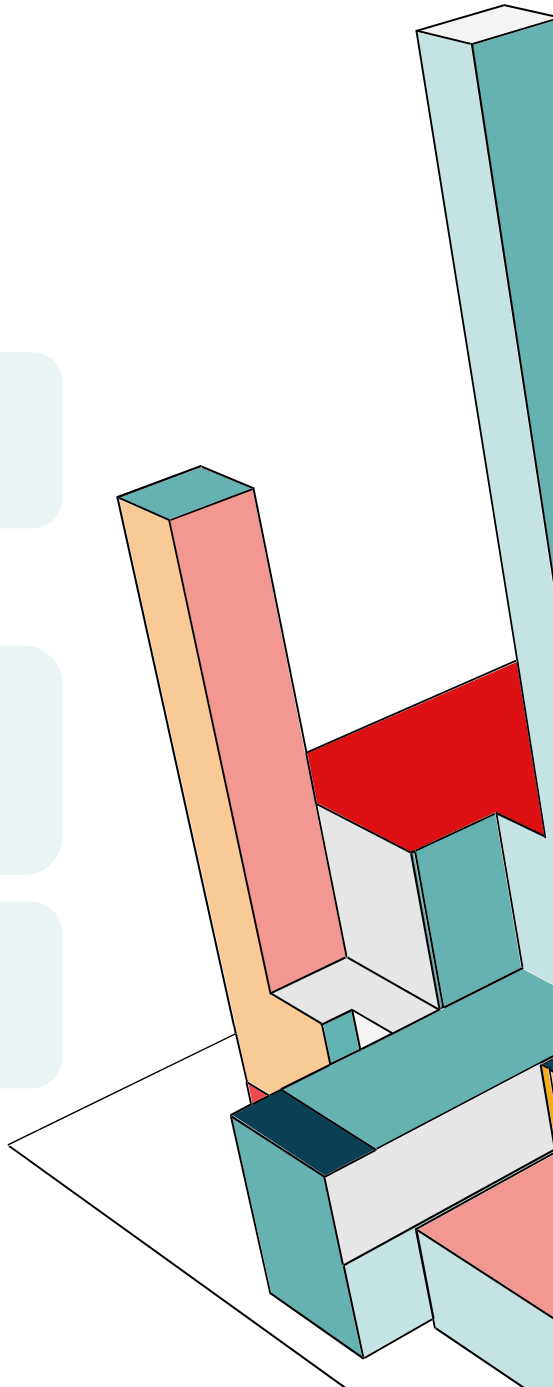
CompCars at a glance

136,726	44,481	1,716
web photos	CCTV photos	car models

Challenges

1. **Hallucinations**
   Edits change parts of the car, miss the hazard or produce illogical images, so every image must be validated.

2. **Domain gap**
   The gap between CompCars photos and real car-wash cameras is unmeasured.

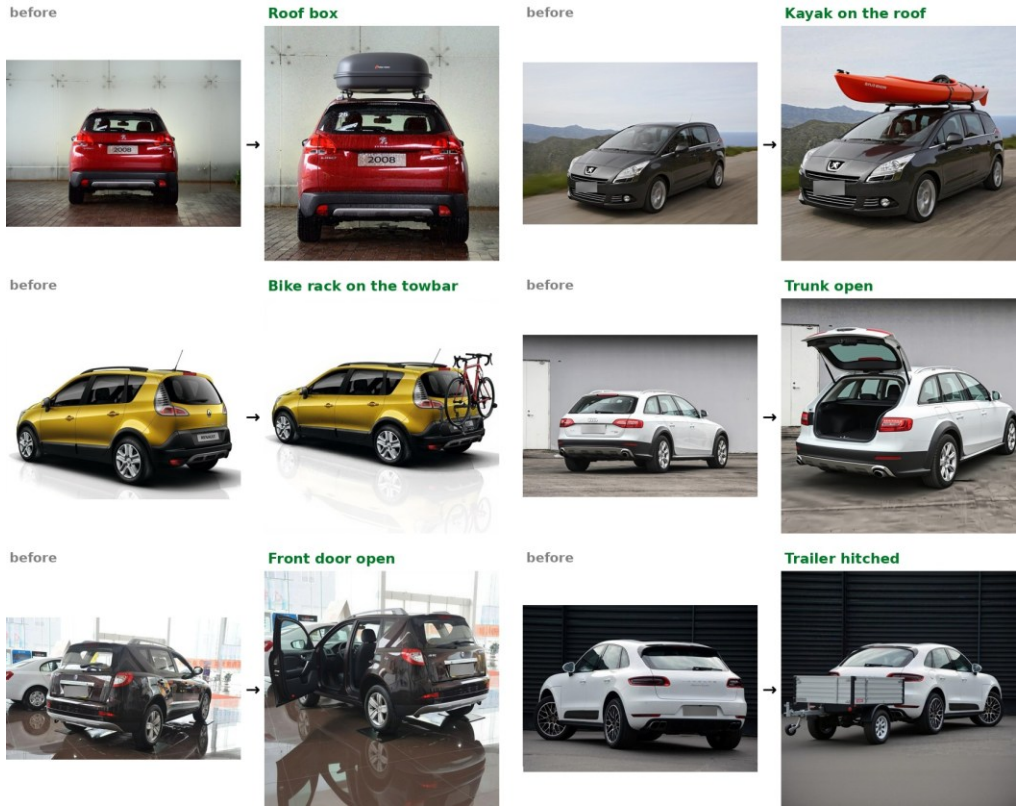


1 Black Forest Labs, FLUX.1 Kontext: Flow Matching for In-Context Image Generation and Editing, arXiv:2506.15742, 2025.

GENERATED EXAMPLES

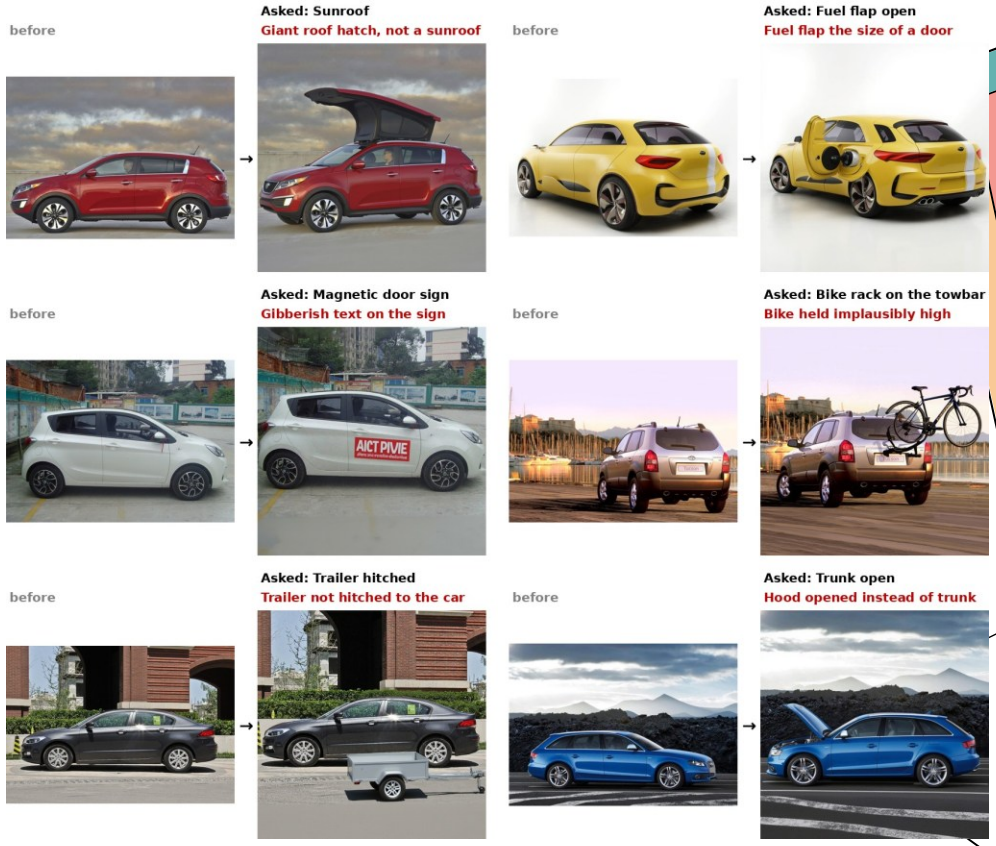
Good edits

One hazard added, rest of the car unchanged



Hallucinations

Wrong part, wrong scale or detached objects



TASK AND SUCCESS METRICS

Scope

- In scope: build the pipeline end to end on CompCars and release the labelled dataset.
- Out of scope: training the final hazard detector. It is a possible extension if time allows.

Hazard classes (23)

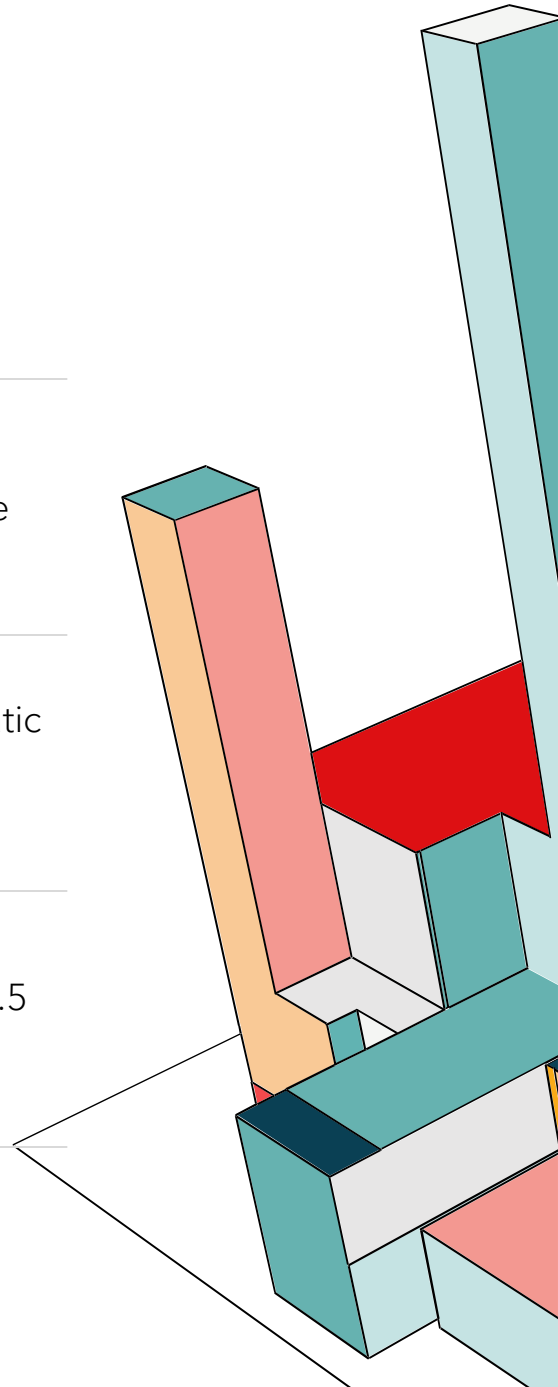
- Openings: doors, trunk, hood, fuel flap, windows, sunroof.
- Roof: roof box, rails, crossbars, bike rack.
- Loads: ladder, kayak, surfboard.
- Rear: towbar rack, exposed towball, trailer and others.

Success metrics

≥50% of each class's images pass validation. Classes below this are dropped.

≥80% agreement between the automatic checks and human review on a sample.

≥90% of SAM 3[1] masks usable: IoU ≥0.5 against a hand-labelled sample.



1 Carion et al. (Meta), SAM 3: Segment Anything with Concepts, arXiv:2511.16719, 2025.

WHY THE DATASET MATTERS

A labelled dataset is enough to train a small detector that runs on the edge



Illustrative example, not a required project output: a lightweight detector (e.g. YOLO) trained on labelled images, running on the car-wash cameras.

The project stops at the labelled dataset; training and deploying the detector come afterwards.

